Dear Dr. Thrall,

Please consider our paper, "Traits predict forest phenological responses to photoperiod more than temperature" for publication as a full paper in *Ecology Letters*.

Climate change is altering the environmental cues of growth globally, producing shifts in the timing of life history events (phenology), altering resource use and producing novel species interactions. But to date, we have observed considerable variation in communities' responses to environmental change, likely reflecting differences in species assemblages and the effects of other species' traits on the timing and patterns of growth. In plants, decades of research has found consistent relationships between traits and growth strategies under different environments—giving rise to frameworks like the leaf or wood economic spectra[1,2]. Within these frameworks, species fall along a gradient of fast and acquisitive to slow and conservative strategies. Phenology was foundational to the theory that underlies these spectra[3], but has rarely been included because it is often challenging to measure and highly variable across species when measured in natural conditions.

Here, we paired two massive controlled environment experiments with trait data for over 1400 individual plants across 47 species and eight sites, testing how phenology fits within major frameworks for plant growth. Based on decades of research into the relationship between traits and plant growth strategies, we explored whether early versus late leafout is correlated with acquisitive versus conservative growth strategies, and how these relationships change across a 6 $^\circ$ latitudinal and 55° longitudinal gradient.

We found evidence that phenology does fit within major frameworks for plant growth, with early to late budburst correlating with plant size and leaf metrics. Of the six traits we tested, four of the traits were related to leafout through photoperiod cues. Despite previous studies finding temperature to be the dominant controller of leafout, we found no support that other traits relationship with temperature cues influence leafout timing. But traits did vary across sites, with most exhibiting strong latitudinal gradients and/or differences between our two transects, likely reflecting differences in winter intensities across the sites.

Our results provide the most compelling evidence of trait-phenology relationships across North America with implications for how forest communities will shift with climate change. Photoperiod is generally considered the weakest cue of leafout, but shared importance to other traits we observed suggests it may pose a stronger constraint on species ability to adapt to warmer spring conditions than previous thought. Accounting for these phenology-trait relationships will improve our ability to accurately forecast future changes in species growth strategies and species interactions.

All authors contributed to this work and approve this version for submission. The manuscript is 4014 words with a 149 word summary, and 4 figures and is not under consideration elsewhere. We hope you find it suitable for publication in *Ecology Letters*, and look forward to your response.

Sincerely,

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[1] Wright, I. J., Westoby, M., Reich, P. B., Oleksyn, J., Ackerly, D. D., Baruch, Z., Bongers, F., Cavender-Bares, J., Chapin, T., Cornellissen, J. H. C., Diemer, M., Flexas, J., Gulias, J., Garnier, E., Navas, M. L., Roumet, C., Groom, P. K., Lamont, B. B., Hikosaka, K., Lee, T., Lee, W., Lusk, C., Midgley, J. J., Niinemets, Ü., Osada, H., Poorter, H., Pool, P., Veneklaas, E. J., Prior, L., Pyankov, V. I., Thomas, S. C., Tjoelker, M. G., and Villar, R. The worldwide leaf economics spectrum. *Nature* **428**, 821–827 (2004).

[2] Chave, J., Coomes, D., Jansen, S., Lewis, S. L., Swenson, N. G., and Zanne, A. E. Towards a worldwide wood economics spectrum. *Ecology Letters* **12**(4), 351–366, April (2009).

[3] Grime, J. P. Evidence for the Existence of Three Primary Strategies in Plants and Its Relevance to Ecological and Evolutionary Theory Author (s): J . P . Grime Source : The American Naturalist , Vol . 111 , No . 982 (Nov . - Dec ., 1977), pp . 1169-1194 Published. *The American Naturalist* **111**(982), 1169–1194 (1977).